

Linux Filesystem Hierarchy

Core idea: Linux organizes files, devices, directories, and system resources into one unified hierarchical tree beginning at the root directory */*, regardless of which physical storage devices or partitions are involved.

Visual Architecture Map

Directory	Primary Purpose
/	Root of the entire filesystem tree
/boot	Kernel and bootloader initialization files
/dev	Device nodes representing hardware and virtual devices
/etc	System and service configuration files
/home	Personal directories for regular users
/root	Home directory of the root administrator
/opt	Optional or third-party application software
/run	Volatile runtime state created during the current boot
/tmp	Temporary short-term files
/var	Variable data such as logs, spools, and web content
/usr	Major collection of programs, libraries, and shared resources
/proc	Virtual view of processes and kernel information
/sys	Virtual view of devices, drivers, and kernel information
/mnt	Administrator-managed temporary mount point
/media	Mount point commonly used for removable media

1. Root and Essential System Infrastructure

/ — The Root Directory

The root directory `/` is the foundation of the entire Linux filesystem tree. Every file, device, and directory branches from it.

- It is the starting point of the filesystem hierarchy.
- The root directory contains the major system directories rather than ordinary user documents.

Key concept: Do not confuse the root directory `/` with the root user's home directory `/root`.

/boot — Static Boot Loader Files

Contains files required during the early stages of system startup and operating-system initialization.

Typical Content	Purpose
Linux kernel (for example, <code>vmlinuz</code>)	Provides the kernel image loaded during boot.
<code>initrd</code> / <code>initramfs</code>	Initial RAM filesystem used during early boot, including required drivers and tools.
GRUB files/configuration	Supports the bootloader and boot process.

Administrative note from the source: `/boot` may be placed on a separate partition on some systems to help preserve bootability when the main filesystem has problems.

/dev — Device Files

A virtual directory containing special device nodes through which software can interact with hardware and other device-like interfaces. Linux provides a file-oriented interface to many devices.

Example	Meaning
<code>/dev/sda</code>	A device node commonly representing a storage disk.
<code>/dev/nvme0n1</code>	A device node commonly representing an NVMe storage device.
<code>/dev/null</code>	Discards data written to it; reading from it immediately produces EOF.
<code>/dev/random</code>	Provides random data from the kernel's random-number subsystem.

2. Executables and Configuration

/bin and /sbin — System Binaries

Directory	Traditional Role	Examples
/bin	Essential command binaries available to users.	ls, cat
/sbin	System-administration binaries used for system maintenance.	fdisk, reboot

Modern Linux: Many distributions use **UsrMerge**, where directories such as **/bin** and **/sbin** are symbolic links to corresponding locations under **/usr**.

/etc — Host-Specific System Configuration

The **/etc** directory is the central location for system-wide configuration files. These files control the behavior of the operating system and installed services.

Path	Purpose
/etc/fstab	Defines filesystems and storage devices that are mounted according to system configuration.
/etc/ssh/sshd_config	Contains configuration for the OpenSSH server and its remote-access behavior.

Source note: The supplied material describes **/etc** as containing static text configuration rather than executable programs.

3. User Data and Runtime Environments

/home — User Home Directories

Contains personal directories for regular users, such as **/home/alice**. Users normally store documents, downloads, and personal configuration here.

- A user normally has read/write access to their own home directory.
- Access to another user's home directory depends on permissions and other access controls.

/root — Root User Home Directory

The personal home directory of the **root** administrator account. It is separate from **/home**.

Why separate? The source material highlights that keeping **/root** directly under the root filesystem can allow an administrator to work on a system even if the filesystem containing **/home** is unavailable.

/run — Ephemeral Runtime Data

Stores volatile runtime information generated since the current system boot.

- Process IDs (PIDs).
- Daemon and service sockets.
- Information about active login sessions.

It is commonly implemented using **tmpfs**, so its contents are recreated when the system boots.

4. Variable and Optional Data

`/var` — Variable Data Files

The `/var` hierarchy contains data that changes during normal system operation and may grow over time.

Path	Purpose
<code>/var/log</code>	System and application logs used for monitoring, troubleshooting, and investigation.
<code>/var/mail</code>	Mail spool data for local users on systems using this traditional layout.
<code>/var/www</code>	A commonly used location for web-server content.

`/opt` — Optional Add-on Software

Reserved for optional or third-party software packages that are installed as relatively self-contained application bundles.

Example from the source: A large third-party application may have its own directory under `/opt`, such as `/opt/google/chrome`.

5. Virtual and Kernel Filesystems

`/proc` and `/sys`

These are virtual filesystems rather than ordinary collections of files stored permanently on a disk. They expose information and interfaces provided by the Linux kernel.

Filesystem	Purpose	Example
<code>/proc</code>	Exposes information about processes and kernel/system state.	<code>/proc/cpuinfo</code> provides processor information.
<code>/sys</code>	Provides a structured view of devices, drivers, and kernel-related system information and interfaces.	Device and driver information under <code>/sys</code> .

Cybersecurity relevance: Understanding `/proc` and `/sys` is useful when investigating processes, system state, devices, drivers, and suspicious activity on Linux systems.

6. Universal Software Resources

/usr — User Utilities and Applications

The **/usr** hierarchy contains a large portion of the operating system's programs, libraries, documentation, and other shared resources. It contains its own hierarchy of subdirectories.

Path	Purpose
/usr/bin	Standard user-space executable programs.
/usr/lib	Libraries and other shared program resources used by software.
/usr/local	Software and other files installed locally by an administrator, commonly outside the distribution's normal package-managed locations.

Important distinction: **/usr/local** is intended for locally installed software and data. The source emphasizes that this separation helps avoid conflicts with ordinary distribution-managed software.

7. Temporary Spaces and Mount Points

/tmp — Temporary Files

A shared location for short-term temporary files created by users and programs.

- Programs can use **/tmp** as a scratch space.
- Temporary data may be removed automatically by the operating system or cleanup mechanisms.

Security note: Because **/tmp** can be writable by many users and processes, permissions and safe temporary-file handling are important when writing security-sensitive software.

/mnt and /media — Storage Mount Points

Directory	Typical Use
/mnt	A conventional mount point for administrators to manually mount temporary filesystems or storage.
/media	A conventional location where removable media such as USB drives may be mounted automatically.

Mounting concept: A filesystem from another storage device can be attached to a directory in the Linux tree. After mounting, its contents become accessible through that directory.

8. Quick Reference

Path	Remember It As
/	Root of everything
/boot	Boot files and kernel
/dev	Device nodes
/etc	Configuration
/home	Regular users' homes
/root	Root user's home
/run	Current boot's runtime state
/tmp	Temporary files
/var	Changing/variable data

Path	Remember It As
/opt	Optional third-party software
/usr	Programs, libraries, and shared resources
/proc	Processes and kernel information
/sys	Devices, drivers, and kernel interfaces
/mnt	Manual mounts
/media	Removable media

Key Takeaway

The Linux filesystem is a unified tree beginning at **/**. Each major directory has a defined role: **/etc** for configuration, **/home** for regular-user data, **/var** for changing data, **/usr** for software resources, **/dev** for device nodes, and **/proc** and **/sys** for kernel-provided system information and interfaces. Knowing this hierarchy makes Linux administration, troubleshooting, and cybersecurity investigation much easier.